

Lesson 5: Applying Color Filters to Images

Using Color Filters to Create 3D Images

Take Two Picture from Slightly Different Angles

When focusing on an object, each of your eyes will view that object from a slightly different angle relative to the background. We can recreate this effect in a flat image using color filters. To do so, we'll need to take two pictures of an object, one for each of our eyes' viewing angles, and apply the appropriate color filter. As you'll see, the difference is very subtle; so you may have to experiment a bit to get two photos that work well together.

Note, since the images will presumably be taken with the same camera and settings, they should both already be the same size. If not, we'd need to resize one of them to match the other.

```
Ileft  = imread("hurlyLeft.jpg"); % view from left eye
Iright = imread("hurlyRight.jpg"); % view from right eye
imshowpair(Ileft,Iright,'montage')
```



Create Two Color Filters

We'll use the `newFilter` function from the [Pixels to Pictures](#) toolbox to create color filters. In order to use these filters they must be the same size as our images (`Ileft` and `Iright` are the same size). We can use the `size` function to determine the size of an image (or array). This function returns an array with values for the size along each dimension.

```
sz = size(Ileft)
```

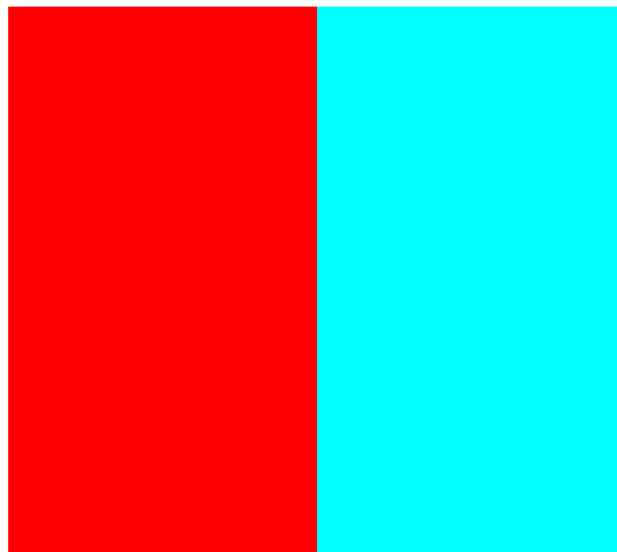
```
sz = 1×3
    3264    1836     3
```

Recall, images have three dimensions: rows, columns, and pages. So the variable `sz` contains the number of rows, the number of columns, and the number of pages, respectively. We can then extract these values using what we previously learned about indexing.

```
rows = sz(1);  
cols = sz(2);
```

We'll set the RGB values of these filters to match the two colors of our 3D glasses: red and cyan (green/blue). Recall we specify colors using Red, Green, and Blue (RGB) values, which range from 0 to 255.

```
red = [255 0 0];  
cyan = [0 255 255];  
R = newFilter(rows,cols,red);  
C = newFilter(rows,cols,cyan);  
imshowpair(R,C,'montage')
```



Use Filters to Remove Opposite Color from Each Image

For most 3D glasses, our left eye will be looking through the red lens and our right eye will be looking through the cyan lens. Consequently, we want to make the left-eye image red toned and the right-eye image cyan toned. We can create a red-toned image by filtering out all the cyan shades; and we can create a cyan-toned image by filtering out all the red shades.

So then, if you put on your 3D glasses and look at these two images separately with one eye closed at a time, your left eye should see mostly just the red-toned image (on the left) and your right eye should see mostly just the cyan-toned image (on the right).

```
Ired = Ileft - C; % remove cyan from left-eye image  
Icyan = Iright - R; % remove red from right-eye image
```

```
imshowpair(Ired,Icyan,'montage')
```



Add Filtered Images Together

Try on your 3D glasses to see if it worked!

```
I3D = Ired + Icyan;  
imshow(I3D)
```

